

MoiraiVaR: VaR-Aware Adaptation of Time-Series Foundation Models for Volatility Forecasting

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Abstract. Point-accurate volatility forecasts do not necessarily support calibrated tail-risk decisions. This paper presents MoiraiVaR, a risk-aware adaptation of pretrained time-series foundation models for multi-horizon volatility forecasting. A frozen Moirai-family backbone encodes a 60-day return window, and a compact regression head predicts volatility at five horizons. Training combines multi-horizon mean squared error with a Student- t Value-at-Risk (VaR) pinball loss, weighted by $\lambda = 0.2$. We evaluate the adaptation in a benchmark containing 1,322,330 predictions from 26 configurations, nine equity markets, and five horizons. The decision protocol combines three point losses, two volatility-dynamics criteria, and four VaR criteria. Each VaR criterion is formed by averaging diagnostics from independently backtested Normal, Student- t , and filtered-historical-simulation thresholds. Under both balanced 50:50 and risk-priority 30:70 accuracy-risk policies, MoiraiVaR–Moirai-MoE ranks first by both Simple Additive Weighting and TOPSIS. It remains first when the accuracy weight varies from 30% to 70% and when the tracking-correlation eligibility threshold increases from 0 to 0.10. The winner is not uniformly best on every criterion: Moirai 2 yields lower point losses, while GJR–GARCH obtains higher VaR pass rates. The evidence therefore supports MoiraiVaR as the strongest aggregate accuracy-risk trade-off in the evaluated model set rather than as a universal winner under every tail model.

Keywords: MoiraiVaR; time-series foundation model; volatility forecasting; Value-at-Risk; multi-criteria decision making

1 Introduction

Volatility forecasts serve two related but non-equivalent purposes. They describe the expected scale of future price movements, and they parameterize tail-risk quantities used in decision support. ARCH and GARCH models encode conditional heteroskedasticity through recursive variance equations [1, 2]; asymmetric specifications additionally capture the unequal response to positive and negative return shocks [3]. Neural sequence models offer greater representational capacity, but low point loss alone does not guarantee realistic volatility amplitude or calibrated Value-at-Risk (VaR). Forecast rankings can also change with the volatility target, proxy, and scoring rule [4, 5].

Recent hybrids combine econometric structure with recurrent or attention-based models [8, 9]. Time-series foundation models extend this direction by transferring representations learned across heterogeneous domains: Moirai provides a universal forecasting encoder [16], while Moirai-MoE routes tokens through sparse experts [17]. Neither design directly represents the asymmetric cost of financial tail violations. Selecting a foundation model only by MSE or MAE therefore leaves the operational risk objective implicit.

This study addresses that mismatch with *MoiraiVaR*, a parameter-efficient adaptation that couples a pretrained Moirai-family representation to a multi-horizon volatility head and a VaR-aware training objective. The volatility term fits five forecast horizons, while a Student- t pinball term penalizes errors in the 1% left-tail threshold. The model is assessed with a separate multi-distribution protocol: Normal, Student- t , and filtered historical simulation (FHS) are back-tested independently, and their method-level diagnostics are averaged before multi-criteria ranking. This separation prevents the distribution used in the training penalty from silently determining the evaluation conclusion.

The paper makes three contributions. First, it specifies a risk-aware adaptation of Moirai, Moirai 2, and Moirai-MoE using a compact volatility head and a joint point-tail objective. Second, it evaluates 26 heterogeneous model configurations with nine criteria spanning point accuracy, volatility dynamics, and VaR calibration across three tail constructions. Third, it shows that the MoiraiVaR-Moirai-MoE configuration is the only candidate ranked first by both SAW and TOPSIS under both 50:50 and 30:70 accuracy-risk policies. The analysis also reports where this conclusion weakens, including distribution-specific ranking changes and modest absolute VaR pass rates.

2 Related Work

GARCH remains a demanding volatility benchmark [6], with FIGARCH extending persistence to long-memory dynamics [7]. Neural hybrids combine this econometric structure with recurrent or attention-based models [8, 9]. MoiraiVaR instead adapts a pretrained representation through a small volatility head and an explicit tail loss.

Transformer forecasting progressed from self-attention [10] to efficient and decomposition-based variants such as Reformer, Informer, Autoformer, and FEDformer [11–14]. Their performance is not uniform across datasets [15]. Moirai and Moirai-MoE address this transfer problem through universal pretraining and sparse expert routing [16, 17]; neither is trained specifically for volatility-derived tail risk.

Normal VaR follows the variance-covariance construction popularized by RiskMetrics [19]. Conditional Student- t innovations model heavier tails [20], while FHS uses an empirical filtered innovation distribution [21]. Kupiec coverage and Christoffersen independence tests assess complementary exception properties [22, 23]. We retain all three tail constructions because each can rank the same volatility forecast differently.

SAW applies additive-utility aggregation [24]; TOPSIS ranks distance from ideal and anti-ideal solutions [25, 26]. Reporting both exposes sensitivity to normalization and compensation.

3 MoiraiVaR

3.1 Foundation representation and volatility head

Let r_t denote the percentage log return at time t , and let

$$\mathbf{x}_t = (r_{t-59}, \dots, r_t) \in \mathbb{R}^{60}$$

be the return context. The implementation left-pads the context to 64 values and partitions it into four patches of length 16. The patches and observation masks are processed by a pretrained Moirai-family encoder. In the parameter-efficient configuration evaluated here, the backbone parameters are frozen and only the downstream head is optimized.

If $\phi_\theta(\mathbf{x}_t)$ is the sequence of backbone representations, temporal mean pooling gives

$$\mathbf{h}_t = \frac{1}{P} \sum_{p=1}^P \phi_\theta(\mathbf{x}_t)_p.$$

A two-layer multilayer perceptron maps $\mathbf{h}_t$ to volatility forecasts for horizons $\mathcal{H} = \{1, 3, 5, 10, 21\}$:

$$\hat{\sigma}_t = W_2 \text{Dropout}(\text{ReLU}(W_1 \mathbf{h}_t + \mathbf{b}_1)) + \mathbf{b}_2.$$

The hidden dimension is 256 and the dropout probability is 0.2. The same head is attached separately to Moirai, Moirai 2, and Moirai-MoE backbones. This construction isolates whether a shared pretrained representation can be redirected toward volatility and risk without retraining the full foundation model.

3.2 VaR-aware joint objective

For target volatility $y_{t,h}$ and prediction $\hat{\sigma}_{t,h}$, the multi-horizon term is

$$\mathcal{L}_{\text{vol}} = \frac{1}{|\mathcal{B}||\mathcal{H}|} \sum_{t \in \mathcal{B}} \sum_{h \in \mathcal{H}} (\hat{\sigma}_{t,h} - y_{t,h})^2.$$

The training-time tail threshold uses the one-step prediction, a zero conditional mean, and the lower Student- t quantile:

$$v_{t,\alpha} = \hat{\sigma}_{t,1} t_{\alpha,\nu}, \quad \alpha = 0.01.$$

With $u_t = r_{t+1} - v_{t,\alpha}$, the pinball term [18] is

$$\mathcal{L}_{\text{VaR}} = \frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} (\alpha - \mathbf{1}\{u_t < 0\}) u_t.$$

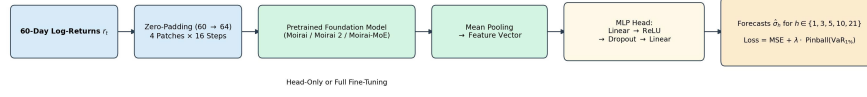


Fig. 1. MoiraiVaR architecture. A 60-day return sequence is padded and patched before encoding by a Moirai-family foundation model. Mean-pooled features feed a compact volatility head with five forecast outputs. Training combines multi-horizon MSE with a weighted 1% VaR pinball loss.

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{vol}} + \lambda_{\text{VaR}} \mathcal{L}_{\text{VaR}}, \quad \lambda_{\text{VaR}} = 0.2.$$

The degrees of freedom are estimated from training-return excess kurtosis κ : $\nu = 30$ when $\kappa \leq 0$, and otherwise $\nu = \text{clip}(4 + 6/\kappa, 2.1, 30)$. The tail term therefore changes with market-level training data while the risk weight remains fixed. The evaluated configuration trains only the regression head with AdamW and leaves the Moirai-family backbone frozen.

4 Accuracy–Risk Evaluation

4.1 Three VaR constructions

Training uses a Student- t penalty, but evaluation does not privilege that distribution. The three constructions follow Normal, conditional Student- t , and FHS practice [19–21]. For $\alpha \in \{0.01, 0.05\}$, predicted scale $\hat{\sigma}_t$, Normal quantile z_α , and Student- t quantile $t_{\alpha, \hat{\nu}}$, the parametric thresholds are

$$\widehat{\text{VaR}}_{\alpha, t}^N = \hat{\sigma}_t z_\alpha, \quad \widehat{\text{VaR}}_{\alpha, t}^t = \hat{\sigma}_t t_{\alpha, \hat{\nu}}.$$

FHS uses the empirical quantile of a shifted 250-observation standardized residual window:

$$\widehat{\text{VaR}}_{\alpha, t}^{\text{FHS}} = \hat{\sigma}_t \hat{Q}_\alpha \{u_{t-250}, \dots, u_{t-1}\}.$$

The one-step shift ensures that the FHS threshold uses only preceding residuals.

Every method $m \in \{N, t, \text{FHS}\}$ is backtested independently within each market-horizon block. Let $PR_{\alpha, m}$ be the proportion of blocks passing both the Kupiec coverage and Christoffersen independence tests [22, 23], and let $E_{\alpha, m} = |\bar{p}_{\alpha, m} - \alpha|$, where $\bar{p}_{\alpha, m}$ is the equally weighted mean block violation rate. The four risk criteria are

$$\overline{PR}_\alpha = \frac{1}{3} \sum_m PR_{\alpha, m}, \quad \overline{E}_\alpha = \frac{1}{3} \sum_m E_{\alpha, m}, \quad \alpha \in \{0.01, 0.05\}.$$

These equations average method-level diagnostics. They do not average the three numerical VaR thresholds.

4.2 Forecast criteria and Sanity Gate

Forecast quality is measured by MSE, MAE, and QLIKE. Volatility dynamics add the amplitude error

$$e_\sigma = |s_{\hat{y}}/s_y - 1|$$

and non-negative tracking error

$$e_\rho = 1 - \max\{\rho(y, \hat{y}), 0\}.$$

Together with the two averaged pass rates and two averaged violation errors, the decision matrix contains nine criteria. A configuration is eligible only if all criteria are finite, $e_\sigma \leq 0.9$, and its mean tracking correlation is positive. This Sanity Gate excludes forecasts that achieve competitive scalar losses while failing to reproduce volatility dynamics.

4.3 Decision policies and statistical analysis

The balanced policy assigns 50% to forecast/dynamics and 50% to VaR; the risk-priority policy assigns 30% and 70%. Criteria receive equal weights within each group. SAW uses benefit/cost min-max normalization and additive aggregation [24]; TOPSIS measures distance to ideal and anti-ideal solutions [25, 26].

5 Experimental Study

5.1 Data and competitors

The source contains 1,322,330 daily prediction rows covering VN-Index, VN30, KOSPI, Nikkei 225, DAX 40, Euronext 100, IBEX 35, SMI, and S&P 500 at horizons $\{1, 3, 5, 10, 21\}$. The 26 configurations form 1,170 configuration-market-horizon blocks. Competitors include four GARCH-family models (including FI-GARCH [7]); twelve Transformer configurations based on Reformer, Informer, and Autoformer [11–13]; Moirai, Moirai 2, and Moirai-MoE; their three VaR-aware adaptations; and four GARCH-/Wavelet-Autoformer hybrids. Five configurations fail the Sanity Gate, leaving 21 candidates for MCDM ranking. Informer (Tier 1) fails the positive-correlation condition; Autoformer (Tier 3) fails the amplitude condition; Reformer (Tier 3), Vanilla Transformer (Tier 3), and Informer (Tier 3) fail both.

5.2 Implementation details

The MoiraiVaR runs use a 60-day input, a five-output head, at most 30 epochs, AdamW with head learning rate 10^{-3} and weight decay 10^{-4} , gradient clipping at 1, and validation early stopping with patience 7. The primary experiment evaluates head-only adaptation at $\lambda_{\text{VaR}} = 0.2$. Two independent MCDM runs produce identical decision matrices and rankings.

Across the 45 market–horizon blocks, the Friedman statistics for MSE, MAE, and QLIKE are 813.10, 840.80, and 878.79, respectively, with 25 degrees of freedom and $p < 10^{-150}$. Kendall’s W equals 0.723, 0.747, and 0.781. These omnibus results establish broad performance differences among configurations; they do not establish pairwise superiority of the MCDM winner [27].

5.3 Main rankings

Table 1 reports every eligible configuration with its complete model label. Under 50:50, MoiraiVaR–Moirai–MoE obtains SAW 0.8426 and TOPSIS 0.8401, ranking first by both methods. Its SAW margin over Moirai 2 is 4.59%, while its TOPSIS margin over MoiraiVaR–Moirai 2 is 8.40%. Under 30:70, the same configuration remains first, ahead of GJR–GARCH by 3.52% in SAW and 1.95% in TOPSIS.

SAW and TOPSIS remain closely aligned: Spearman rank correlation is 0.981 ($p = 6.84 \times 10^{-15}$) under 50:50 and 0.877 ($p = 1.87 \times 10^{-7}$) under 30:70. The winner is therefore unchanged even though several lower-ranked candidates move substantially when risk receives more weight.

The winner is a compromise rather than a metric-wise dominator. Table 2 compares it with the strongest pure-accuracy and pure-pass-rate alternatives. Moirai 2 has the lowest MSE, MAE, and QLIKE, and the highest tracking correlation. GJR–GARCH has the highest 1% and 5% VaR pass rates. MoiraiVaR–Moirai–MoE remains near the top of the accuracy criteria while improving the 5% pass rate relative to the unadapted Moirai–MoE from 0.304 to 0.393. Its 1% and 5% absolute violation errors also decline from 0.00829 to 0.00786 and from 0.01036 to 0.01008.

The SAW score decomposition makes the balance explicit. Under 50:50, MoiraiVaR–Moirai–MoE receives 0.4907 from the forecast/dynamics block and 0.3518 from the VaR block. Under 30:70, the components become 0.2944 and 0.4926. GJR–GARCH obtains the larger 30:70 risk component (0.6241), but its accuracy component is only 0.1362. MoiraiVaR thus avoids the severe accuracy sacrifice associated with optimizing pass rates in isolation.

5.4 Robustness of the balance claim

MoiraiVaR–Moirai–MoE remains first by both methods at accuracy weights 0.30, 0.40, 0.50, 0.60, and 0.70. It also remains first when the minimum tracking-correlation threshold increases from 0 to 0.05 and 0.10, which reduces the eligible set from 21 to 16 and 13 configurations. These checks show that its consensus rank is not produced by one weight split or by weakly dynamic competitors near the Gate boundary.

Tail-model sensitivity is stronger. Under Normal-only criteria, FI-GARCH is the consensus winner; under Student- t -only criteria, GARCH–Autoformer Tier 2 ranks first; and under FHS-only risk priority, GJR–GARCH ranks first. For MoiraiVaR–Moirai–MoE itself, the 1% pass rates are 0.556, 0.156, and 0.200 for FHS, Normal, and Student- t . At 5%, they are 0.511, 0.578, and 0.089. The averaged primary analysis rewards a configuration that performs comparatively

Table 1. Complete MCDM ranks for all 21 eligible configurations. Models are ordered by their balanced-policy performance. The best model is shown in bold.

Configuration	50:50 policy		30:70 policy	
	SAW rank	TOPSIS rank	SAW rank	TOPSIS rank
MoiraiVaR–Moirai-MoE ($\lambda = 0.2$)	1	1	1	1
MoiraiVaR–Moirai 2 ($\lambda = 0.2$)	3	2	6	6
Moirai 2	2	4	3	9
Moirai-MoE	4	3	5	7
Moirai	5	6	4	8
MoiraiVaR–Moirai ($\lambda = 0.2$)	6	5	7	5
FI-GARCH	8	7	8	3
GJR-GARCH	7	10	2	2
GARCH-LSTM-Hybrid	9	9	10	10
GARCH	10	8	9	4
GARCH-Autoformer (Tier 1)	11	11	17	11
GARCH-Autoformer (Tier 2)	12	12	15	12
Wavelet-Autoformer (Tier 1)	13	14	12	15
Autoformer (Tier 1)	14	13	11	13
Autoformer (Tier 2)	15	15	13	14
Wavelet-Autoformer (Tier 2)	17	16	16	16
Vanilla Transformer (Tier 1)	16	18	14	18
Reformer (Tier 1)	18	17	18	19
Reformer (Tier 2)	19	19	19	17
Vanilla Transformer (Tier 2)	20	20	20	20
Informer (Tier 2)	21	21	21	21

well across these conflicting constructions, but it does not imply invariance to the choice of tail model.

6 Discussion and Limitations

The main result supports a specific form of risk-aware transfer. The Moirai-MoE representation is retained, while a small head translates its token-level mixture representation into horizon-specific volatility. The pinball term then changes the preferred solution within the neighborhood of high-accuracy foundation models. Relative to the unadapted Moirai-MoE, the VaR-aware version gives up 2.0% in MSE and 4.0% in MAE but gains 8.89 percentage points in the 5% averaged pass rate. The MCDM outcome interprets this exchange as favorable under both decision policies.

The result also illustrates why a single performance label is insufficient. “Best forecast” could refer to point loss, tracking, exception frequency, or a composite decision policy. Moirai 2 is the point-accuracy leader; GJR–GARCH is the pass-rate leader; and MoiraiVaR–Moirai-MoE is the multi-distribution compromise. Reporting these roles separately prevents a composite rank from obscuring the raw evidence.

Several limitations bound the conclusion. First, MCDM scores are relative to the 21 eligible configurations and depend on normalization over that set.

Table 2. Raw aggregate metrics for representative configurations. Lower is better for losses and violation error; higher is better for correlation and pass rate. Bold entries identify the best value in each column.

Configuration	MSE	MAE	QLIKE	ρ	$\overline{PR}_{.01}$	$\overline{E}_{.01}$	$\overline{PR}_{.05}$	$\overline{E}_{.05}$
MoiraiVaR–Moirai–MoE	0.02526	0.10149	0.00932	0.899	0.304	0.00786	0.393	0.01008
Moirai–MoE	0.02476	0.09756	0.00913	0.898	0.296	0.00829	0.304	0.01036
Moirai 2	0.02275	0.09062	0.00912	0.916	0.296	0.00781	0.289	0.01016
GJR–GARCH	0.19112	0.29196	0.05141	0.444	0.348	0.00623	0.526	0.01114

Second, equal averaging across Normal, Student- t , and FHS is transparent but normative, and FHS has a shorter effective sample because of its 250-observation warm-up. Third, the primary experiment evaluates only $\lambda_{\text{VaR}} = 0.2$; a controlled multi- λ ablation is needed to identify the adaptation frontier. Fourth, aggregated market-horizon metrics may hide local reversals, so the first consensus rank does not imply a win in every market or horizon. Fifth, the MCDM result is a deterministic decision analysis rather than a confidence interval for pairwise superiority. Future work should combine purged temporal evaluation, positive output parameterization, paired block-level effect sizes, and leave-one-market-out sensitivity.

7 Conclusion

MoiraiVaR connects a pretrained time-series foundation model to financial tail-risk requirements through a multi-horizon volatility head and a VaR-aware joint objective. Among 26 evaluated configurations, MoiraiVaR–Moirai–MoE with $\lambda_{\text{VaR}} = 0.2$ is ranked first by SAW and TOPSIS under both balanced and risk-priority policies. Its rank is stable across weight and eligibility-threshold perturbations because it retains near-leading forecast quality while improving VaR calibration over the unadapted Moirai–MoE. Distribution-specific rankings and absolute backtest pass rates remain heterogeneous. The evidence therefore supports MoiraiVaR as the strongest observed aggregate accuracy–risk trade-off in this benchmark and motivates controlled ablation of risk-aware foundation model adaptation.

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